BORING LOG

BH-LA-01

Project: Elk Basin Dam PFR Investigation
Location: Left Abutment Crest, Sta. 2+40
Ground Elev: 6,238.5 ft
Drilling Method: HSA to 72 ft / HQ wireline core 72-120 ft

MWDE Project No. 2025-DAM-0034
Date Drilled: August 18-21, 2025
Total Depth: 120 ft
Driller: Idaho Core Drilling, Inc.

Groundwater: 42.0 ft bgs (Aug 18)

Instruments: VW piezometer at 95 ft

Depth (ft)	USCS	N / Rec / RQD	Material Description	Samples & Remarks
0–25	GP-GM	N=32 N=38 N=45	GRAVEL FILL — Embankment shell zone. Well-graded gravel with sand and silt, angular to subangular, dry to moist, dense to very dense. Placed fill.	SS-1 to SS-6 (5-ft intervals)
25–45	CL	N=12 N=14 N=13	CLAY — Embankment core zone. Lean clay, brown, moist to wet, stiff to very stiff. LL=41-44, PI=19-21. Homogeneous, no evidence of piping or erosion channels. Intact.	SS-7 to SS-13 ST-1 at 33-35 ft (Shelby tube)
GW				
45–65	GP	N=28 N=35 N=42	GRAVEL FILL — Embankment downstream shell / filter zone. Poorly graded gravel, clean, angular, moist, dense. Filter layer at 60-62 ft (sand, SP).	SS-14 to SS-18
65–72	SM/ML	N=22 N=28	ALLUVIAL DEPOSITS — Original ground surface. Sandy silt to silty sand, brown, moist, medium dense. Thin (0.5 in) clay gouge at 65.0 ft (stripped surface).	SS-19 to SS-20 Foundation contact at 65.0 ft
72–85		Rec: 65% RQD: 18%	WEATHERED GRANODIORITE (GRUS) — Highly to moderately weathered, friable to moderately hard. Sandy decomposition texture, relict foliation. UCS 1,820-4,100 psi. Grout remnants at 74-76 ft.	HQ core RC-1 UCS: W1, W2, W3 Estimated k: 1×10■■ cm/s
85–120 TD		Rec: 94-96% RQD: 78-85%	FRESH GRANODIORITE — Cretaceous biotite granodiorite, gray, coarse-grained, massive to slightly jointed. 2 joint sets (45° and 70°). UCS 21,200-25,600 psi. Hard, competent bedrock.	HQ core RC-2, RC-3 UCS: U1-U4 ITS: T1-T5 VW piez at 95 ft (18.2 psi initial)

BORING LOG

BH-RA-01

Project: Elk Basin Dam PFR Investigation | MWDE Project No. 2025-DAM-0034
Location: Right Abutment Mid-Slope, Sta. 6+80 | Date Drilled: August 22-25, 2025
Ground Elev: 6,210.3 ft | Total Depth: 95 ft
Drilling Method: HSA to 55 ft / HQ wireline core 55-95 ft | Driller: Idaho Core Drilling, Inc.

Groundwater: 35.0 ft bgs (Aug 22)

Instruments: VW piezometer at 75 ft; Inclinometer to 90 ft

Depth (ft)	Graphic	USCS	N / Rec / RQD	Material Description	Samples & Remarks
0–15		GP-GM	N=35 N=40	GRAVEL FILL — Embankment upstream shell. Well-graded gravel with sand and silt, angular, moist, very dense. Placed fill.	SS-1 to SS-3
15–32		CL	N=11 N=13 N=14	CLAY — Embankment core zone. Lean clay, brown, wet, stiff to very stiff. LL=39-41, PI=19-20. Intact, homogeneous, no piping indicators.	SS-4 to SS-10 ST-1 at 22-24 ft
32–48 GW		GP	N=30 N=38	GRAVEL FILL — Embankment downstream shell. Poorly graded gravel, angular, moist, dense to very dense.	SS-11 to SS-13
48–55		GC	N>70 (refusal)	GLACIAL TILL — Very dense, heterogeneous clayey gravel with cobbles and boulders. Gray-brown, moist. LL=26-28, PI=11-12. Subrounded to subangular clasts. Matrix-supported. Creep behavior confirmed in triaxial.	SS-14, SS-15 UD-1 at 50-52 ft (undisturbed tube)
55–70			Rec: 58% RQD: 12%	WEATHERED GRANODIORITE — Completely to highly weathered, soil-like to friable. Gradational contact with till at 55 ft. UCS 1,280-3,800 psi. Sandy grus texture.	HQ core RC-1 UCS: W1-W3
70–95 TD			Rec: 91-93% RQD: 72-82%	FRESH GRANODIORITE — Gray, coarse-grained, massive to slightly jointed. UCS 18,400-23,500 psi. 3 joint sets. Tight joints with thin calcite coating.	HQ core RC-2, RC-3 UCS: U1-U4 ITS: T1-T4 VW piez at 75 ft Incl. to 90 ft

BORING LOG

BH-AC-01

Project: Elk Basin Dam PFR Investigation | MWDE Project No. 2025-DAM-0034
Location: Arch Crest at Crown, Sta. 4+60 | Date Drilled: August 26-27, 2025
Ground Elev: 6,245.0 (dam crest) ft | Total Depth: 28 ft
Drilling Method: NX continuous diamond core (0-28 ft) | Driller: Idaho Core Drilling, Inc.

Groundwater: Seepage at lift joints (7, 12 ft depth)

Instruments: None

Depth (ft)	Graphic	USCS N / Rec / RQD	Material Description	Samples & Remarks
0 – 4		Rec: 92% RQD: 65%	MASS CONCRETE — Severe ASR zone. Extensive map cracking, gel deposits in cracks and pores, reaction rims on rhyolite aggregate. White to amber gel. f'c = 4,850 psi. f't = 320 psi.	NX core C1 (UCS) ST1 (split tensile) Petro specimen 1
4 – 8		Rec: 95% RQD: 78%	MASS CONCRETE — Moderate ASR. Reaction rims on aggregate, minor gel at crack intersections. Cracking less extensive. f'c = 5,920 psi. Seepage at lift joint at 7 ft.	C2 (UCS) Petro specimen 2
8 – 14		Rec: 97% RQD: 88%	MASS CONCRETE — Mild ASR. Trace reaction rims on few aggregate particles. No gel, no cracking. Seepage at lift joint at 12 ft. f'c = 6,480 psi.	C3 (UCS) ST2 (split tensile) Petro specimen 3
14 – 22		Rec: 98% RQD: 92%	MASS CONCRETE — Sound. No ASR indicators. Original mix design (Type II cement, rhyolite coarse agg., natural sand fine agg.). f'c = 6,720-6,890 psi. f't = 510 psi. Concrete/rock contact at 22.0 ft — intact bond, sound contact grout.	C4, C5 (UCS) ST3 (split tensile) Petro specimen 4
22 – 28		Rec: 97% RQD: 92%	FRESH GRANODIORITE — Foundation rock. Gray, coarse-grained, massive, excellent quality. UCS 24,200 psi. ITS 1,690 psi. No weathering, tight joints.	RC-1 U1 (UCS) T1 (ITS)

28 **TD**